MODULE 32 · DOMAIN 5 - IMPLEMENT INFORMATION EXTRACTION SOLUTIONS

Content Understanding and document extraction

OCR, layout and field extraction; analyzer families and their model requirements; markdown for RAG; grounding; and choosing between Content Understanding and Document Intelligence.

DOMAIN 5 - Information extraction WEIGHT 10-15% MODULE 32 of 39

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OCR **reads** the page. Layout **maps** it. Fields **answer questions** about it. Most pipelines need all three.

From a messy file to grounded content

01
Read
OCR text, including scans.

02
Map
Paragraphs, tables, figures.

03
Structure
Markdown or JSON output.

04
Extract
Fields defined by your schema.

05
Ground
Confidence and source spans.

06
Use
Index it, or hand it to an agent.

ON THE EXAM

One analyzer can do all of it, or you can stop after any step, depending on what downstream needs.

Analyzer families

FAMILY	EXAMPLES	USE FOR
Content extraction	prebuilt-read, prebuilt-layout	Text, tables, structure, no models needed
Search and RAG	prebuilt-documentSearch and friends	Markdown ready for chunking and indexing
Domain prebuilts	invoice, receipt, ID documents, contracts	Known document types, no labelling
Utility	Schema generation, field discovery	Finding what's in a document type
Custom	Your own field schema	Fields nobody else has defined

ON THE EXAM

Prebuilt analyzers can also be used as tools in agentic flows, and customised further.

Which analyzers need model deployments

- **prebuilt-read and prebuilt-layout use no LLM or embedding models.** No deployments needed.
- **Search analyzers** need a small chat model and an embedding model deployed.
- **Invoice and receipt analyzers** need a larger chat model, plus embeddings.
- Deployments are configured once at the **resource level**, by model name.
- Different extraction operations bill differently, so the cheapest analyzer that fits wins.

> THE SIGNAL
>
> If a scenario says no models are deployed, or cost must stay minimal, read and layout are still available.

What prebuilt-layout gives you

- **Words, paragraphs, tables and figures**, with their positions.
- Document **structure**: sections and formatting.
- **Hyperlinks** embedded in the document.
- Structure is what lets chunking follow headings instead of cutting mid-table.
- Document metadata, such as author and title, depends on the analyzer and API version.

> ON THE EXAM
>
> A table split across two chunks is worse than useless. Layout is what prevents that.

Clean representations for retrieval and agents

- **prebuilt-documentSearch** returns markdown: paragraphs, tables and **figure descriptions** inline.
- Markdown is the shape **chunkers and embedding models prefer**, with headings intact.
- Semantic chunking keeps related content together, as module 29 showed.
- The **MarkItDown** integration picks the analyzer per file type, and emits extracted fields as front matter.
- Figure descriptions mean diagrams contribute text, without a separate verbalization step.

> ON THE EXAM
>
> 'Clean, grounded representations' in the study guide means exactly this: markdown plus grounded fields.

Files to markdown, with fields

```
md = MarkItDown(cu_endpoint=endpoint)

report = md.convert("report.pdf")     # uses prebuilt-documentSearch
webinar = md.convert("meeting.mp4")   # uses prebuilt-videoSearch

invoices = MarkItDown(cu_endpoint=endpoint,
                      cu_analyzer_id="my-invoice-analyzer")
print(invoices.convert("invoice.pdf").markdown)
# ---
# fields:
#   VendorName: CONTOSO LTD.
#   InvoiceDate: '2019-11-15'
# ---
```

ON THE EXAM

Fields arrive as front matter above the markdown.

Fields, confidence and grounding

- A **field schema** names each field, its type and description, as module 23 showed.
- Methods: **extract** verbatim, **generate** from understanding, or **classify** into set values.
- The GA API returns **confidence scores** and **source grounding** for fields.
- Grounding lets a reviewer jump to the exact place a value came from.
- Typed values, such as dates and numbers, are normalised to a canonical format.

> ON THE EXAM
>
> Confidence plus grounding is what makes automated extraction reviewable, and therefore usable in regulated work.

Content Understanding **forces grounding**. If it can't anchor a value in the document, it returns **nothing** rather than a guess.

Content Understanding or Document Intelligence?

CONTENT UNDERSTANDING

Describe what you want

- Zero-shot: describe fields in plain language
- Generalises across templates and languages
- Markdown and grounded fields for RAG and agents
- Add labelled samples later to improve

DOCUMENT INTELLIGENCE

Train on what you have

- Prebuilt models for common forms
- Custom models need at least five samples per variant
- Consistent outputs, confidence and grounding
- Strong for deterministic, structured extraction

THE TIEBREAKER

Microsoft's guide: prebuilt analyzer if one covers your document type; custom analyzer for highly varied templates.

Requirement to approach

REQUIREMENT	APPROACH
Plain text and tables, no models deployed	prebuilt-layout
Markdown for a RAG pipeline	prebuilt-documentSearch
Invoices, receipts, ID documents	Matching domain prebuilt analyzer
Hundreds of supplier templates, no labels	Custom analyzer, fields in plain language
One stable form, labelled samples available	Document Intelligence custom model
Documents inside an agent conversation	Content Understanding context provider
Documents inside a search pipeline	Content Understanding skill in the skillset

ON THE EXAM

Ask three questions: which document type, how varied, and what the output feeds.

Documents in an agent conversation

- The Agent Framework **context provider** analyses attachments and injects the results.
- It works for **documents, images, audio and video**, and removes the binary after analysis.
- For **large documents**, it can upload the markdown to a **file search vector store** instead.
- That keeps the context window small while the content stays searchable.
- Analyzers can also be called as tools, so a planner reads a document when it needs to.

> ON THE EXAM
>
> A 200-page PDF in the prompt is a token bill. The same PDF as searchable markdown is a retrieval problem, solved.

Extraction symptoms and causes

SYMPTOM	LIKELY CAUSE
Fields come back missing	Values not groundable; check scan quality and schema wording
Tables split awkwardly in chunks	Chunking ignoring layout structure
Analyzer fails to run	Required model deployments missing at the resource
Costs higher than expected	A model-backed analyzer where read or layout would do
Same PDF and DOCX give different results	Documented behaviour differs by document type
Great OCR, poor field accuracy	Field descriptions too vague to extract against

ON THE EXAM

Field descriptions are prompts. Vague descriptions produce vague extraction, or nothing at all.

> A public sector team must extract full text and tables from thousands of scanned PDFs, into their own archive format.
>
> Their Foundry resource has no model deployments, and procurement rules make adding model capacity slow and expensive.

Which analyzer should they use?

A prebuilt-documentSearch, for markdown with tables and figure descriptions.

B prebuilt-layout, which extracts text, tables and structure.

C A custom analyzer with a field schema describing each table column.

D prebuilt-invoice, because the archive contains many financial documents.

◆ PAUSE AND DECIDE

Which analyzer should they use?

A prebuilt-documentSearch.
Needs a chat model and an embedding model deployed.

B prebuilt-layout.
Text, tables and structure, with no LLM or embedding models required.

C Custom analyzer.
Field extraction needs models, and they want full text anyway.

D prebuilt-invoice.
A domain analyzer for invoices, needing models, not a full-text extractor.

THE SIGNAL

'No model deployments' rules out everything except read and layout.

> A team is building RAG over technical manuals full of tables and diagrams. Their current pipeline extracts raw text, and chunks split tables in half.
>
> They want chunks that keep tables intact and include something searchable for each diagram.

What should they use?

A. prebuilt-documentSearch, for markdown with tables and figures.

B. prebuilt-read, then their existing fixed-size chunking with more overlap.

C. A Document Intelligence custom model trained on five manuals per variant.

D. An image analysis skill over every page rendered as a picture.

◆ PAUSE AND DECIDE

What should they use?

A prebuilt-documentSearch.
Markdown with intact tables and figure descriptions, the shape chunkers prefer.

B read plus more overlap.
Raw text with no structure. Overlap duplicates the problem, it doesn't fix it.

C Document Intelligence custom model.
Trains for field extraction on known variants, not RAG-ready content.

D Image analysis per page.
Expensive, and loses the document's text structure entirely.

THE SIGNAL

'Tables intact and diagrams searchable' is what the search analyzers were built to produce.

> A finance team receives invoices from 400 suppliers, in wildly different layouts and several languages. They need vendor, total, due date and purchase order number.
>
> They have no labelled training data, and need results within weeks.

What should they build?

A. A Document Intelligence custom model, with five labelled samples per layout.

B. prebuilt-read, then regular expressions to find totals and dates.

C. prebuilt-layout, with manual review of every extracted table.

D. A Content Understanding analyzer describing the fields in plain language.

◆ PAUSE AND DECIDE

What should they build?

A DI custom model per layout.
Five samples per variant across 400 layouts is a labelling project, not a deadline.

B read plus regex.
Brittle across layouts and languages, with no confidence or grounding.

C layout plus manual review.
Extracts structure, then puts a person back in every invoice.

D Content Understanding analyzer.
Zero-shot fields described in plain language, generalising across templates and languages.

THE TIEBREAKER

Microsoft's own guidance names this case: highly varied templates, no labelling, add samples later to improve.

> A team extracts contract fields with Content Understanding. On poor-quality scans, several fields come back missing entirely, while an older OCR service still returned text for those pages.
>
> Managers ask why the service 'fails' where the old one worked.

What's the right explanation and response?

A It's a defect: raise a support case and keep the old OCR service meanwhile.

B Confidence thresholds are too high, so lower them to accept weaker values.

C Grounding is enforced: improve scans, use layout for full text.

D Switch to agentic mode, which reasons past poor image quality to fill fields.

◆ PAUSE AND DECIDE

What's the right explanation and response?

A Report a defect.

Documented behaviour, not a fault: it won't return ungrounded values.

B Lower thresholds.

The values weren't extracted at all, so there's nothing to accept.

C Grounding enforced; improve input, use layout for text.

Missing means ungrounded. Better scans, or a text-first path where full text is the goal.

D Agentic mode.

Adds reasoning over evidence, not evidence that isn't legible.

THE SIGNAL

A missing field is the service refusing to guess. That's a feature in regulated work, and a scan-quality signal.

Document extraction in five lines

- **OCR reads, layout maps, fields answer.** One analyzer can do all three.
- **read and layout need no model deployments.** Search and domain analyzers do.
- **documentSearch gives markdown** with tables and figure descriptions, ready for chunking.
- **Grounding is enforced:** an ungroundable field comes back missing, not guessed.
- **Varied templates, no labels: Content Understanding.** Stable forms with samples: Document Intelligence.

> ON THE EXAM
>
> Domain 5 is complete. Next, module 33 begins the exam skills modules, starting with older answers versus current ones.